F18CE2 - MV Calculus and Real Analysis B: Solutions 6

1 f is differentiable at a if the limit

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

exists.

(a) Using

$$f(a+4h) - f(a-h) = (f(a+4h) - f(a)) + (f(a) - f(a-h)),$$

we get

$$\lim_{h\to 0} \frac{f(a+4h)-f(a-h)}{h} = \lim_{h\to 0} \frac{f(a+4h)-f(a)}{h} + \lim_{h\to 0} \frac{f(a)-f(a-h)}{h}.$$

Since f is differentiable at a,

$$\lim_{h\to 0} \frac{f(a+4h)-f(a)}{h} = 4\lim_{h\to 0} \frac{f(a+4h)-f(a)}{4h} = 4f'(a)$$

$$\lim_{h\to 0} \frac{f(a)-f(a-h)}{h} = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h} = f'(a)$$

$$\implies \lim_{h\to 0} \frac{f(a+4h)-f(a-h)}{h} = 5f'(a).$$

(b) Using

$$f(a+5h) - f(a+2h) = (f(a+5h) - f(a)) - (f(a+2h) - f(a)),$$

we get

$$\lim_{h\to 0} \frac{f(a+5h)-f(a+2h)}{h} = \lim_{h\to 0} \frac{f(a+5h)-f(a)}{h} - \lim_{h\to 0} \frac{f(a+2h)-f(a)}{h},$$

$$\lim_{h\to 0} \frac{f(a+5h)-f(a)}{h} = 5\lim_{h\to 0} \frac{f(a+5h)-f(a)}{5h} = 5f'(a),$$

$$\lim_{h\to 0} \frac{f(a+2h)-f(a)}{h} = 2\lim_{h\to 0} \frac{f(a+2h)-f(a)}{2h} = 2f'(a),$$

$$\implies \lim_{h\to 0} \frac{f(a+5h)-f(a+2h)}{h} = 3f'(a).$$

2 By definition,

$$\lim_{h\to 0^-} \frac{f(3+h)-f(3)}{h} = \lim_{h\to 0^-} \frac{|h|}{h} = \lim_{h\to 0^-} \frac{-h}{h} = -1.$$

$$\lim_{h\to 0^+} \frac{f(3+h)-f(3)}{h} = \lim_{h\to 0^+} \frac{|h|}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

These one sided limits are different, so

$$\lim_{h\to 0} \frac{f(3+h)-f(3)}{h}$$

does not exist.

3 By definition,

$$\lim_{h\to 0^-} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^-} \frac{h^3}{h} = \lim_{h\to 0^-} h^2 = 0.$$

$$\lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^+} \frac{h^2}{h} = \lim_{h\to 0^+} h = 0.$$

These one sided limits are equal, so

$$\lim_{h\to 0} \frac{f(0+h)-f(0)}{h}$$

exists and f is differentiable at 0.

4 Using $g(a) = 0$,

$$\frac{g(a+h)-g(a)}{h} = \frac{hf(a+h)}{h} = f(a+h).$$

Since f is continuous at a, the limit

$$\lim_{h\to 0} f(a+h) = f(a)$$

exists. Hence,

$$\lim_{h\to 0} \frac{g(a+h)-g(a)}{h} = f(a),$$

and so g is differentiable at a.

5 Putting $x = y = 0$ in $f(x+y) = f(x) + f(y)$ we obtain $f(0) = 2f(0)$ so $f(0) = 0$.

Suppose that f is differentiable at 0 and let $a \in \mathbb{R}$. Then

$$\frac{f(a+h)-f(a)}{h} = \frac{f(h)-f(0)}{h}.$$

Since f is differentiable at 0, $\lim_{h\to 0} \frac{f(h)-f(0)}{h} = f'(0)$ so $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists and f is differentiable at a.

6 Putting $x = y = 0$ in $f(x+y) = f(x)\,f(y)$ we obtain $f(0) = f(0)^2$ so $f(0) = 1$ since $f(0) \neq 0$.

Suppose that f is differentiable at 0 and let $a \in \mathbb{R}$. Then

$$\frac{f(a+h)-f(a)}{h} = \frac{f(a)(f(h)-1)}{h} = \frac{f(a)(f(h)-f0))}{h}.$$

Since f is differentiable at 0, $\lim_{h\to 0} \frac{f(h)-f(0)}{h}$ exists, so $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists, and so f is differentiable at a.